

QUEENSTOWN, New Zealand

WMO: 938310

Lat: 45.0175S

Lon: 168.7400E

Elev: 357

StdP: 97.11

Time Zone: 12.00 (NZL)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-4.2	-3.1	-6.6	2.2	-1.9	-5.5	2.5	-0.5	11.5	4.5	9.7	5.2	2.9	40	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	25.4	15.3	23.9	14.9	22.4	14.4	17.0	22.7	16.1	21.3	15.3	20.4	5.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.8	11.0	18.8	14.0	10.4	18.2	13.1	9.8	17.6	48.9	22.8	46.4	21.2	44.1	20.4	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.0	7.9	-5.5	28.5	1.0	1.6	-6.2	29.6	-6.8	30.5	-7.3	31.4	-8.1	32.5		
(5)			-6.1	18.5	1.0	0.9	-6.8	19.2	-7.4	19.7	-7.9	20.2	-8.7	20.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	15.8	15.6	13.4	9.9	7.2	4.3	3.5	5.2	7.7	9.8	11.7	14.1		
	DBStd	4.98	2.92	2.88	2.70	2.71	2.93	2.72	2.78	2.49	2.58	2.65	3.00	2.83		
	HDD10.0	789	1	1	4	34	95	172	201	149	77	37	15	3		
	HDD18.3	3127	89	84	154	253	344	422	459	406	318	266	198	133		
	CDD10.0	724	180	157	109	31	9	1	0	2	9	30	67	129		
	CDD18.3	20	9	7	1	0	0	0	0	0	0	0	1	2		
	CDH23.3	191	99	58	4	0	0	0	0	0	0	0	5	24		
	CDH26.7	19	13	6	0	0	0	0	0	0	0	0	0	1		
(14) Wind	WSAvg	3.4	3.9	3.7	3.3	3.1	2.8	2.8	2.8	2.9	3.3	3.7	4.0	3.8		
(15) Precipitation	PrecAvg	836	80	60	62	61	87	76	64	72	63	72	67	74		
	PrecMax	1217	234	134	177	167	159	161	144	290	184	162	272	178		
	PrecMin	634	24	12	24	9	30	12	8	6	9	25	1	6		
	PrecStd	135	38	32	31	40	39	42	31	53	45	34	60	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	27.4	23.9	20.0	17.1	13.9	12.8	14.6	17.9	20.4	23.9	25.7		
		MCWB	16.3	16.9	15.0	12.8	11.0	8.8	7.8	8.3	9.9	11.9	14.3	14.7		
	2%	DB	25.8	25.0	22.0	18.0	15.4	12.1	11.1	12.6	15.7	18.4	21.3	23.7		
		MCWB	15.5	15.5	14.3	12.0	10.4	7.9	7.1	7.7	9.2	11.1	13.2	14.4		
	5%	DB	24.1	23.5	20.5	16.5	13.9	10.6	9.7	11.2	14.3	16.8	19.5	22.1		
		MCWB	14.9	15.0	13.8	11.2	9.8	7.3	6.5	7.0	8.7	10.3	12.2	14.0		
	10%	DB	22.3	22.0	19.1	15.3	12.5	9.4	8.4	10.0	13.0	15.4	17.8	20.3		
		MCWB	14.4	14.5	13.2	10.7	9.0	6.7	5.7	6.5	8.2	9.6	11.4	13.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	18.7	17.0	14.1	12.4	9.8	8.6	8.9	11.1	12.9	15.8	16.9		
		MCDB	24.6	25.6	21.1	18.2	15.3	12.6	10.9	12.9	15.2	18.1	21.1	22.3		
	2%	WB	16.9	17.1	15.8	12.8	11.3	8.6	7.6	8.2	10.0	11.8	14.2	15.7		
		MCDB	23.1	22.8	19.7	16.2	14.0	10.9	10.3	11.7	14.1	16.8	19.1	21.2		
	5%	WB	16.1	16.0	14.7	12.0	10.3	7.7	6.8	7.4	9.2	10.9	13.1	14.8		
		MCDB	21.6	21.0	19.1	15.4	13.1	10.2	9.4	10.6	13.2	15.7	18.0	19.8		
	10%	WB	15.2	15.1	13.8	11.1	9.3	6.9	5.9	6.7	8.5	10.1	12.1	14.0		
		MCDB	20.5	20.4	18.3	14.6	12.1	9.2	8.2	9.6	12.4	14.5	16.9	19.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	10.8	10.1	9.3	8.0	7.7	7.9	8.4	9.2	9.6	10.0	10.3		
		MCDBR	14.4	13.7	11.9	10.9	9.7	9.3	9.1	9.8	11.1	11.8	12.8	13.5		
		MCWBR	6.1	5.9	6.0	6.4	6.1	6.3	6.2	6.1	6.3	6.2	6.2	5.9		
		MCDBR	11.6	11.7	9.9	9.5	8.6	8.4	8.4	8.7	9.9	10.6	11.2	11.3		
	5% WB	MCWBR	5.5	5.5	5.7	6.1	6.0	6.1	6.0	5.8	6.0	6.0	5.9	5.6		
(40) Clear-Sky Solar Irradiance	taub		0.317	0.311	0.295	0.286	0.277	0.267	0.262	0.279	0.298	0.307	0.309	0.310		
	taud		2.569	2.605	2.659	2.649	2.629	2.585	2.603	2.604	2.542	2.534	2.560	2.597		
	Ebn at Noon		1005	984	956	890	821	788	829	883	933	977	1008	1019		
	Edh at Noon		102	93	80	67	56	52	56	69	88	99	103	101		
(44) All-Sky Solar Radiation	RadAvg		6.00	5.40	4.02	2.66	1.57	1.22	1.44	2.17	3.28	4.49	5.63	6.14		
	RadStd		0.44	0.43	0.24	0.22	0.13	0.14	0.11	0.16	0.30	0.34	0.45	0.45		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (5 neighbors)		N/A	N/A	+0.78	N/A	N/A	N/A	N/A	N/A	N/A
		+0.27	N/A	+0.77	N/A	N/A	-47	-112	N/A	N/A

Nomenclature: See separate page